Water storage trends and attribution credibility

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Research objectives

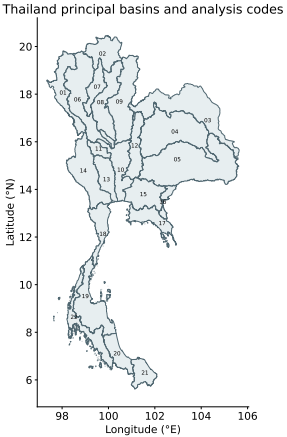
- Storage trends for all 22 principal basins and national context.
- Climate-driven storage reconstruction, paired human effects and groundwater evidence.
- Interactive decision support with reproducible code and data.

Proposed research contribution

- Separate robust national decline from product-dependent basin rankings.
- Test climate memory beyond seasonality and native forcing.
- Audit human attribution through model discrepancy and well-network sensitivity.

Integrated empirical contribution; no claim of a first-ever algorithm or journal acceptance.

Study domain



ONWR principal-basin geometry. Basin codes and full names are retained in the thesis.

Documented inputs and distinct scientific roles

Record	Period	Role
CSR RL06.3	2002–2024	Observed TWSA
GLDAS + G3P	2002–2023/24	Storage compartments
WaterGAP pairing	1981–2023	Naturalized/historical experiment
CHIRPS and cube	2003–2024	Forcing and context
511 Thai wells	Mostly annual	Groundwater consistency

Full provenance, checksums and special-month audit accompany the data.

Groundwater storage balance

$$GWSA = TWSA - SMSA - SWSA - CWSA - SWEA$$

- All compartments use mm water equivalent and a common reference.
- Identical valid cells preserve the regional accounting identity.
- Product depth and filtering differences remain structural uncertainty.

GLDAS Noah and G3P compartment data; Sharifi et al. (2025), doi:10.5194/hess-29-6985-2025.

Three meanings of evaluation

Comparison	What it tests
Emulator / naturalized target	Reconstruction of a parent model
Matched land storage / GLDAS	Cross-model component consistency
Residual / groundwater heads	Groundwater pattern agreement

A strong score in one comparison does not validate the other two.

Chronological climate evaluation

Role	Years	Use
Warm-up	1981–1990	Antecedent weather memory
Train	1991–2009	Fit and internal tuning
Validation	2010–2015	Freeze ensemble choices
Test	2016–2023	Held-out performance

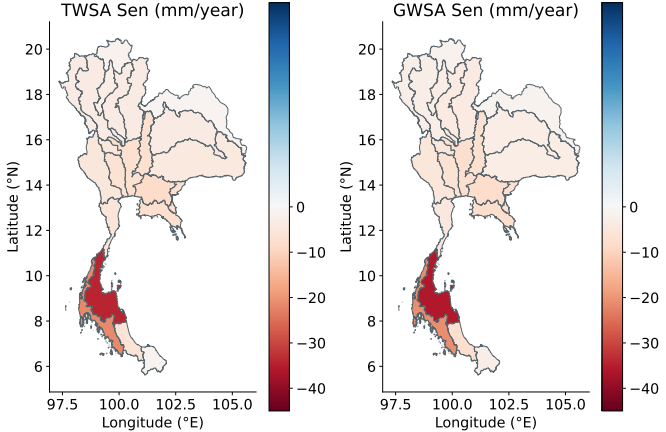
No GRACE lag, well-head feature or unrestricted year trend enters the primary emulator.

National CSR-based storage decline

Storage	Months	Sen mm/year	95% interval
TWSA	237	−7.51	−9.79 to −5.68
GWSA	225	−6.84	−8.77 to −5.11

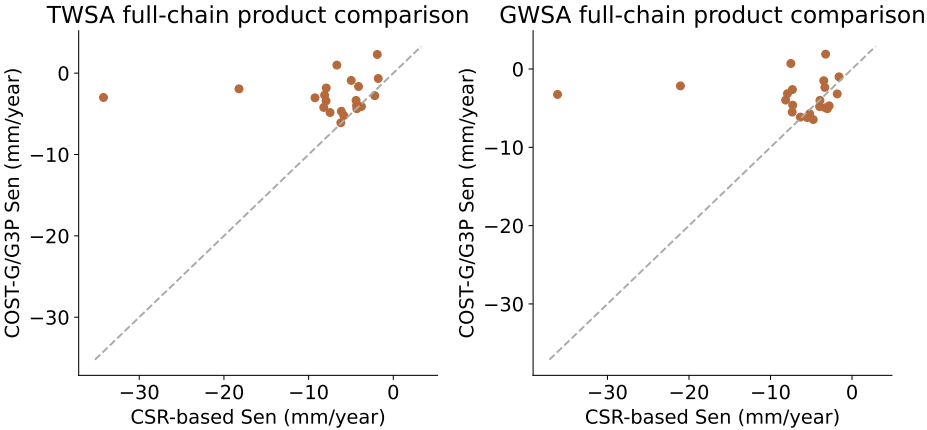
TWSA ends September 2024; GWSA ends September 2023. Intervals are conditional within-product estimates.

Basin storage trends



Basin-level screening, not single-grid-point CSR inference. All 22 basins retain the provider caution.

Product-dependent basin rankings



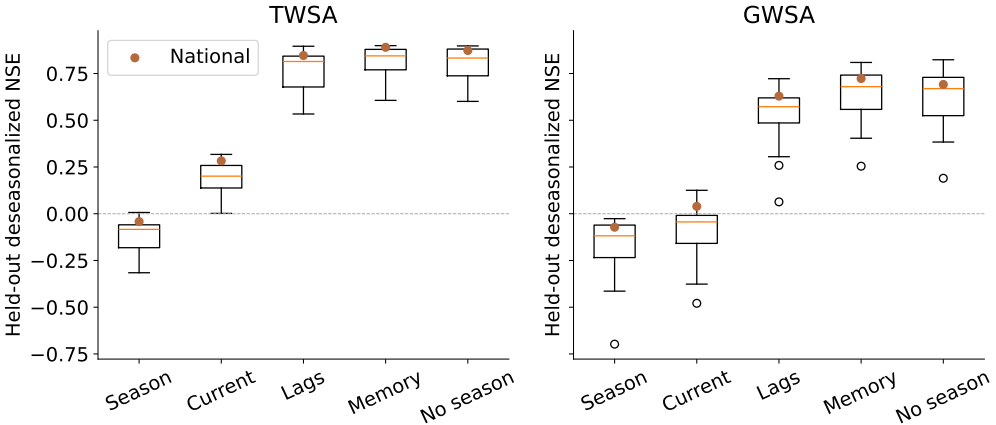
Spearman rank agreement: TWSA 0.138; GWSA 0.027. Common months and cells; GWS chains also differ in compartments.

Spatial support is shared

- National support touches 68 native mascons; inverse concentration is about 50.1.
- All 22 principal basins are smaller than the provider’s approximate caution area.
- Native concentration and overlap are diagnostics, not independent sample counts.

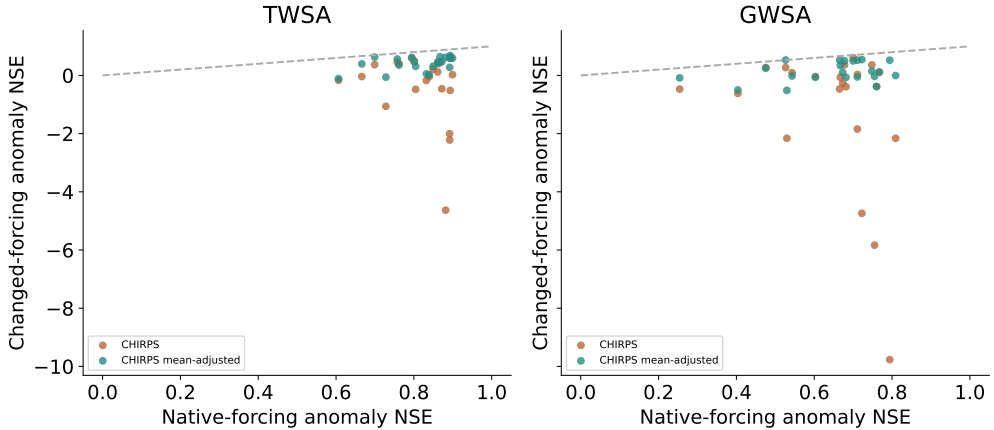
CSR RL06.3 native mapping and basin intersection weights; provider documentation.

Climate memory improves anomaly reconstruction



National anomaly NSE: current → memory is 0.283 → 0.890 for TWSA and 0.039 → 0.723 for GWSA.

Independent precipitation changes model skill



CHIRPS replacement: anomaly NSE becomes 0.688 and 0.562. This is forcing sensitivity, not rainfall-accuracy validation.

Held-out observed rainfall response

Storage	P accumulation	Delay	Test r	Paired months
TWSA	12 months	1 month	0.680	79
GWSA	12 months	2 months	0.606	76

Selected in 2003–2015; tested in 2016–2023. Empirical delay is not aquifer residence time.

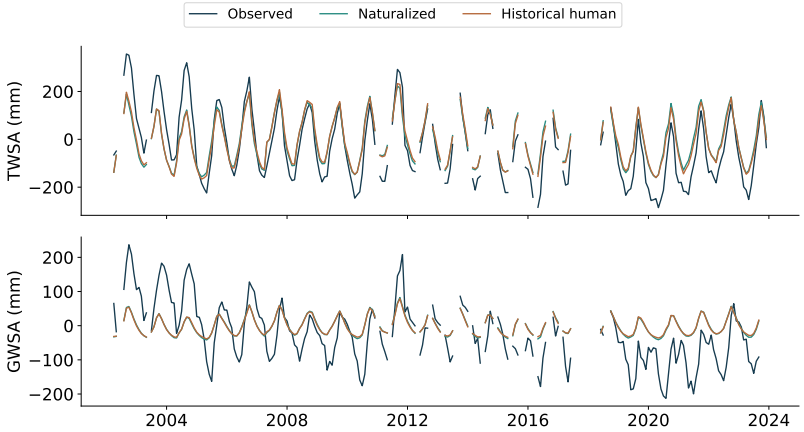
Attribution retains unexplained discrepancy

$$R = O - C = (A - C) + (O - A) = H + E$$

- O: observed; C: naturalized; A: historical-human model.
- H = A − C is the human perturbation represented by the model.
- E = O − A remains unresolved observation–model discrepancy.

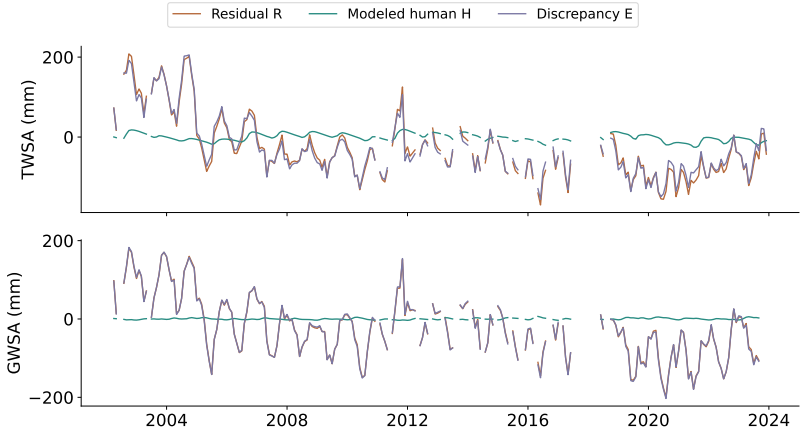
All terms share observed support and the 2004–2009 anomaly reference.

Observed and modeled storage



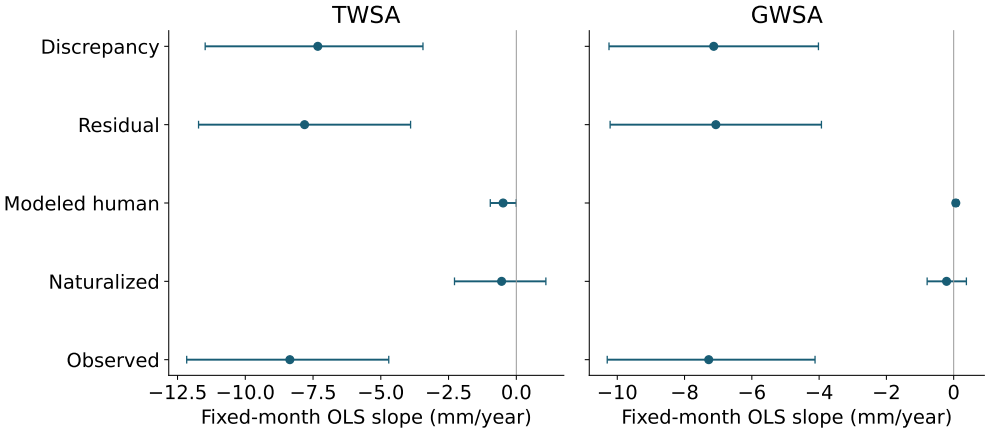
CSR observations and paired WaterGAP storage; no invented 2024 counterfactual.

Residual and paired human effect differ



R = H + E exactly. The residual alone is not a measured pumping history.

Additive national trend partition



OLS monthly fixed effects, identical dates and 999 shared two-year-block multipliers. Sen slopes are not additive.

Discrepancy magnitude diagnostic

Storage	D	Conditional interval
TWSA	0.938	0.822–0.997
GWSA	0.991	0.978–0.999

$D = |\beta E| / (|\beta H| + |\beta E|)$. A descriptive magnitude index, not a causal human fraction.

Groundwater observations

- 5,484 real aggregates from 511 wells; 506 annual wells and five monthly wells.
- Co-located wells are grouped before regional averaging.
- Heads remain in head units; aquifer storage coefficients are not invented.

GROW dataset, doi:10.5281/zenodo.15149480; source flags and original intervals retained.

Panel and period control change validation

Panel	TWS r	GWS r	GWS years
All overlap	0.127	0.037	17
Fixed eligibility/wider history	0.657	0.511	17
Common period	0.325	0.029	8
Strict balanced	0.305	-0.000	8

2013–2022 common-period comparisons retain only eight usable storage years. No mission-gap interpolation.

Groundwater evidence remains limited

- Strict balanced groundwater residual–head correlation is approximately zero.
- 382 eligible annual wells reduce to 225 coordinate-station trend trajectories.
- 54.2% decline in head, with a station-resampling interval of 47.1–60.0%.

Head-direction results do not estimate groundwater mass or validate total human TWSA.

Conclusions

- Broad storage decline is more robust than local severity rankings.
- Climate memory reconstructs a naturalized model well; attribution remains conditional.
- Groundwater panel tests do not validate the whole residual as human-caused depletion.

Decision support and reproducibility

- Inspect basin records, products, conditional uncertainty and observation support.
- Download source-specific inputs, scientific tables and executable scripts.
- github.com/pratikkhanal16/thailand-water-storage-dss

The extended CSR study is versioned separately from the earlier release.

Research priorities

- Aquifer-specific monthly heads, screened intervals and verified storage coefficients.
- Independent pumping, recharge and reservoir-operation histories.
- Additional naturalized models and physically harmonized compartment products.

Selected references

- Save et al. (2016): doi:10.1002/2016JB013007
- Müller Schmied et al. (2024): doi:10.5194/gmd-17-8817-2024
- Sharifi et al. (2025): doi:10.5194/hess-29-6985-2025

Complete bibliography and primary source metadata are included in the thesis and manuscript.

Related Thai work and contribution

- Abhishek et al. (2021): doi:10.1016/j.jhydrol.2021.126868
- Sharma et al. (2025): doi:10.3390/rs17101731
- Contribution: joint product, climate-memory, discrepancy and observation-network audit.

No claim that GRACE analysis, storage subtraction or groundwater validation is new to Thailand.

Validation beyond groundwater levels

Independent evidence	What it can test
Reservoir volume and operation	Surface-water contribution and management timing
Pumping and consumptive use	Withdrawal mechanism; returns and recharge still matter
InSAR / GNSS subsidence	Aquifer deformation; requires geomechanical context

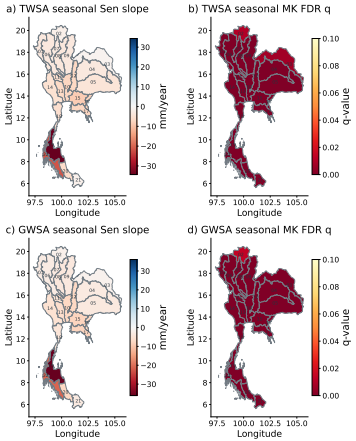
No single proxy directly measures total human-only storage. Avoid using the same data in subtraction and validation.

New independent reservoir check

Product	Matched months	Adjusted TWS r
Scene-supported Level 1	73	0.439 [0.188, 0.665]
Provider-interpolated Level 2	228	0.381 [0.144, 0.552]

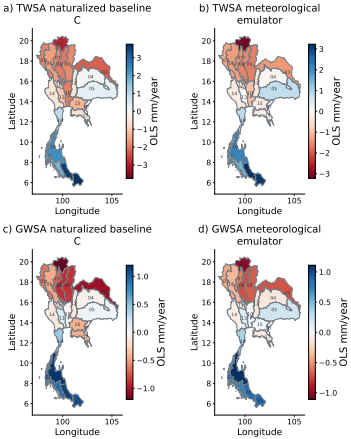
26 pre-2004 reservoirs; calendar-month and linear-trend effects removed on matched support. Conditional year-cluster intervals. MSEA-Res: doi:10.5194/essd-17-2693-2025.

Spatial GRACE storage trends and Mann Kendall evidence



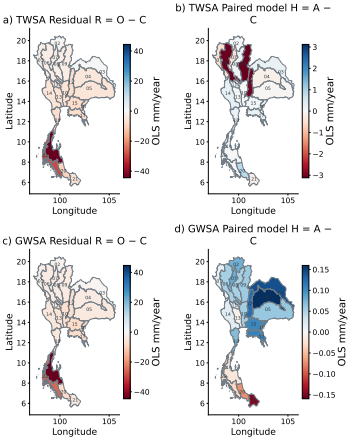
CSR basin-average trends; seasonal MK does not adjust serial dependence. All basins retain provider support cautions.

Spatial naturalized storage and climate reconstruction



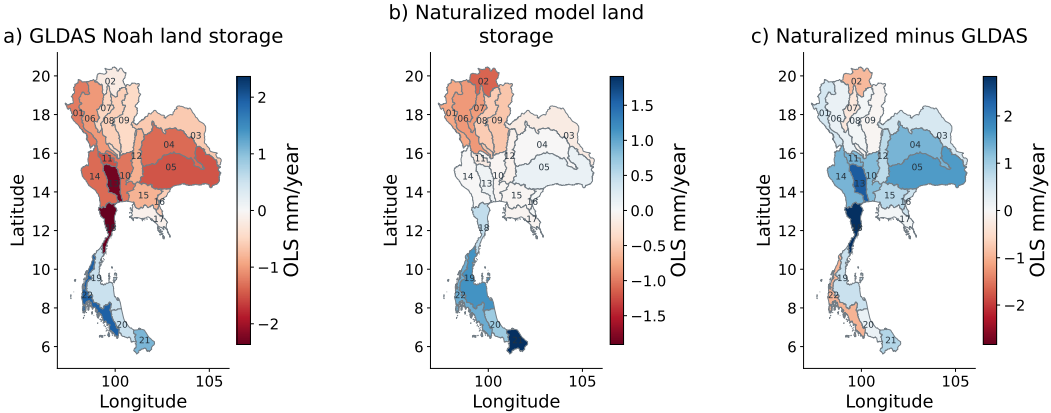
Naturalized baseline C and statistical emulator are related model estimates; neither is an observed counterfactual.

Spatial residual and modeled human perturbation



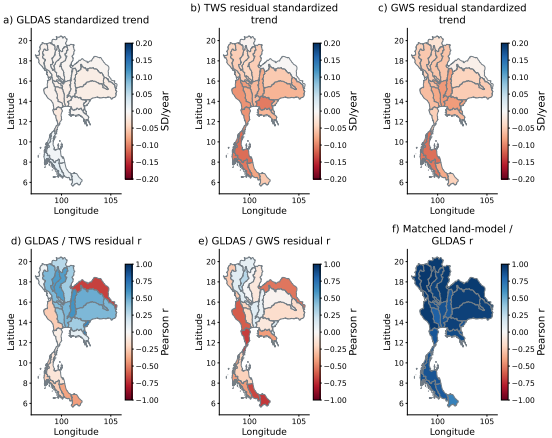
R = O − C = H + E; each panel has its own scale. H is the represented model perturbation, not pumping observations.

Spatial GLDAS matched compartment comparison



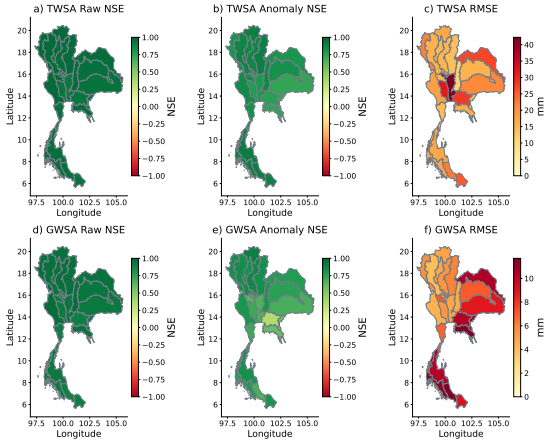
GLDAS and naturalized land storage both include soil, canopy and snow; no groundwater or surface water.

Spatial GLDAS and residual comparisons



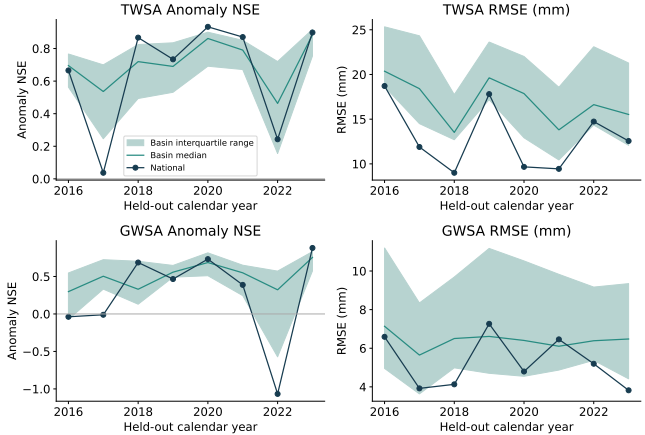
Different compartments: standardized timing/shape comparisons, not a human-only validation target.

Spatial climate model evaluation



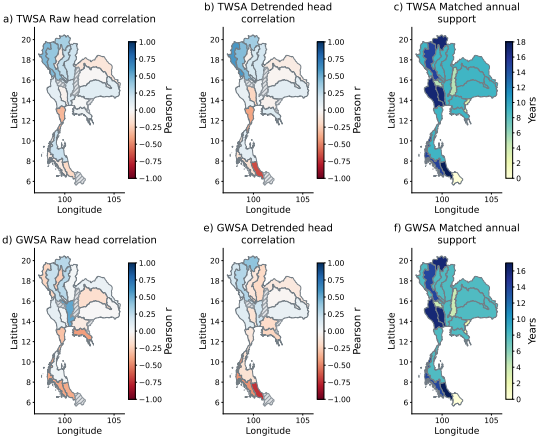
Frozen ensemble; 2016–2023 parent-model verification. NSE colours saturate below −1.

Climate model evaluation through time



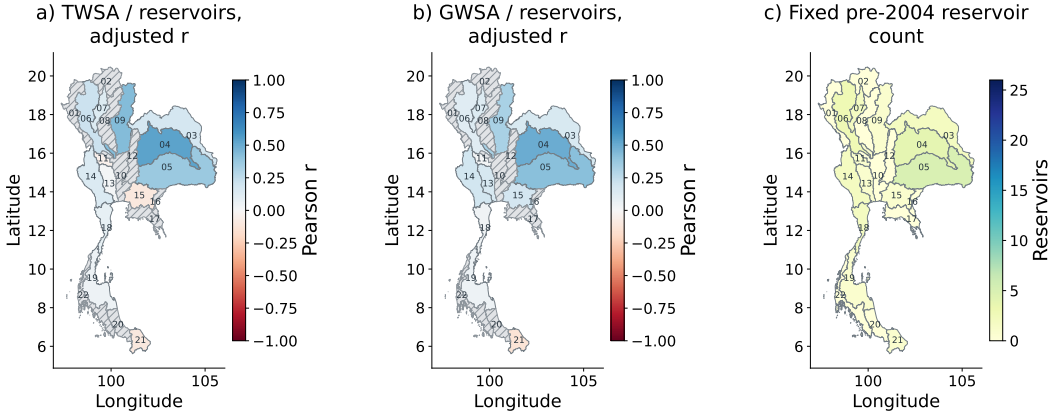
Annual diagnostics use 12 modeled months; spatial IQR is not a confidence interval.

Spatial evaluation of residual agreement with groundwater



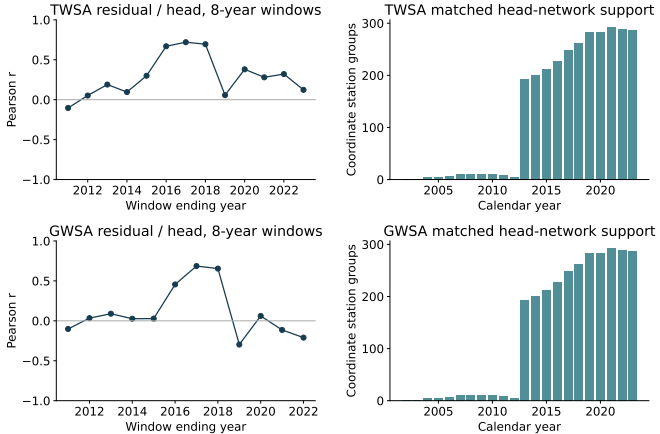
Head correlations and annual support; hatching means insufficient evidence, not zero effect.

Spatial independent reservoir component evidence



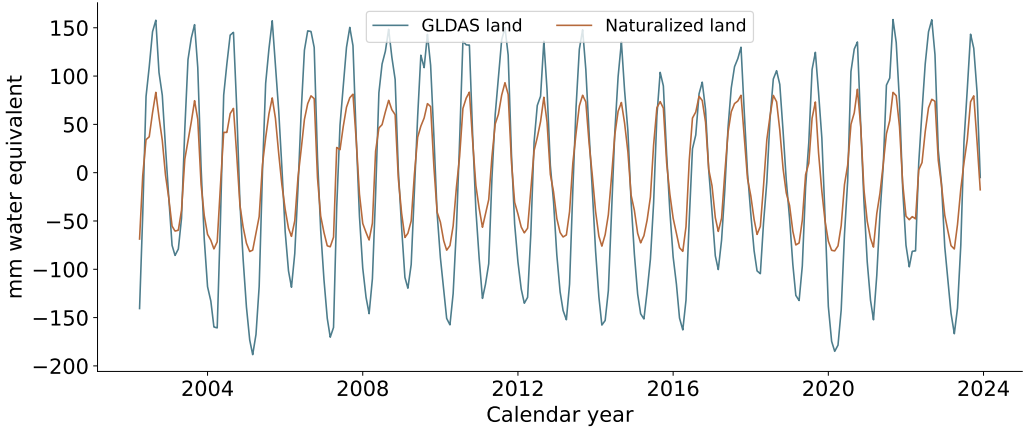
Independent MSEA-Res scene estimates; reservoir changes respond to both climate and operations.

Residual validation metrics and network support through time



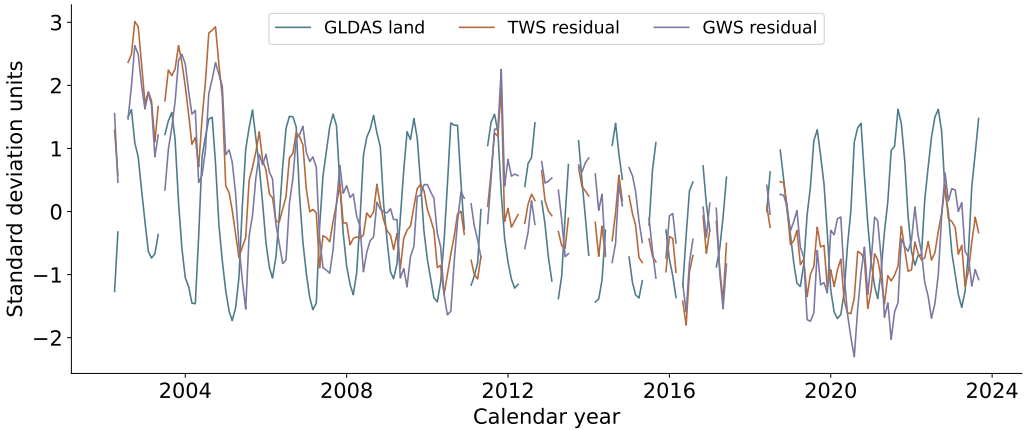
Overlapping 8-year windows require six annual pairs. Station counts are not temporal sample sizes.

Matched GLDAS comparison through time



National soil + canopy + snow comparison. The revised report and manuscript include temporal subplots for all 22 basins.

GLDAS and residual timing comparison



Separately standardized on joint observed months. GLDAS land storage does not represent human-only residual ground truth.